

HW 4

1

Find all of the essential prime implicants for the following Boolean functions.

a

$$F = y_1'(y_0' + x_0 + x_1) + x_1(x_0 + y_0')$$

✓ Answer ✓

$$= y_1'y_0' + y_1'x_0 + y_1'x_1 + x_1x_0 + x_1y_0'$$

$y_0y_1 \backslash x_0x_1$	00	01	11	10
00	1	1	1	1
01	0	1	1	0
11	0	0	1	0
10	0	1	1	1

All 5 of these terms are essential and cannot be expanded, therefore are prime implicants.

$y_0y_1 \backslash x_0x_1$	00	01	11	10
00	1	1,3,5	1,2,3,4,5	1,2
01	0	5	4,5	0
11	0	0	4	0
10	0	3	2,3,4	2

b

$$F = x_1'x_0'y_1'y_0' + x_1'x_0y_1'y_0 + x_1x_0y_1y_0 + x_1x_0'y_1y_0'$$

✓ Answer

$y_0y_1 \backslash x_0x_1$	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

By visual inspection, these are already essential prime implicants

c

$F = w'x'z' + w'x'y + wxz + wx'y'$

✓ Answer

$wx \backslash yz$	00	01	11	10
00	1		2	1,2
01				
11		3	3	
10	4	4		

From this we can see the first prime implicant is not prime, as $y'z'x'$ would cover the essential node from it.

We can also see that $y'zw$ can cover the other node of the fourth prime implicant.

Therefore, our only prime implicants are $w'x'y$ and wxz

d

$F = y'z + wyz + w'(y' + z')$

✓ Answer

$wx \backslash yz$	00	01	11	10
00	1	1		1
01	1	1		1
11		1	1	
10		1	1	

From the table, we can see that only zw and $z'w'$ are prime implicants.

2

Using the tabulation method, find all the prime implicants for the following Boolean expressions.

a

$F = w'x' + w'xy + wx + wx'y' + xy$

✓ Answer

$F = \Sigma(0, 1, 3, 4, 6, 7)$

1s	Minterm	Binary	2	3
0	Trevor Nichols	000	00_, 0_0	0_2 / 10

1s	Minterm	Binary	2	3
1	1, 4	001, 010	0_1, _10	_1_
2	3, 6	011, 110	_11, 11_	
3	7	111		

w' and y are essential prime implicants.

b

$$F = yz + w'z' + wy'z$$

✓ **Answer**

$$F = \Sigma(0, 2, 3, 5, 7)$$

1s	Minterm	Binary	2
0	0	000	0_0
1	2	010	01_
2	3, 5	011 101	_11 1_1
3	7	111	

$$w'z' + w'y + yz + wz$$

c

$$F = x'y'z' + wy'z + xyz + w'yz'$$

✓ **Answer**

1s	Binary	2
0	0000	_000, 00_0
1	1000, 0010	100_, 0_10
2	1001, 0110	1_01, 011_
3	1101, 0111	11_1, _111
4	1111	

$$x'y'z' + w'x'z' + wz'y' + w'yz' + wy'z + w'xy + wxz + xyz$$

d

$$F = wy'z' + wx'y' + wy + w'x'y'z + wxyz'$$

✓ Answer

1s	Binary	2	3
0			
1	1000 0001	1__0 100_ 10_0 _001	1__0 10__ _001
2	1100 1001 1010	11_0 10_1 1_10 101_	1_1_
3	1011 1110	1_11 111_	
4	1111		

$$wz' + wx' + x'y'z + wy$$

3

Find all the minimal covers for the following Boolean functions.

a

$$F = \Sigma(0, 4, 5, 8, 9, 10, 12, 13, 14, 15)$$

✓ Answer

1s	Binary	2	3
0	0000	0__0 _000	__00
1	0100 1000	010_ _100 100_ 10_0 1_00	_10_ 1_0_ 1__0
2	0101 1001 1010 1100	_101 1_01 1_10 110_ 11_0	11__
3	1101 1110	11_1 111_	
4	1111		

Prime implicants

$$c'd' + bc' + ac' + ad' + ab$$

	$c'd'$ __00	bc' _10_	ac' 1_0_	ad' 1__0	ab 11__
0000	E				
0100	e	e			
1000	e		e	e	
0101		E			
1001			E		
1010				E	
1100	e	e	e	e	e
1101		e	e		e
1110				e	e
1111					E

Essential Prime implicants and cover

$$c'd' + bc' + ac' + ad' + ab$$

b

$$F = \Sigma(0, 1, 2, 4, 5, 6, 9, 11, 13, 15)$$

✓ Answer

1s	Binary	2	3
0	0000	000_ 00_0 0_00	0_0_ 0__0
1	0001 0010 0100	0_01 _001 0_10 010_ 01_0	__01
2	0101 0110 1001	_101 10_1 1_01	1__1
3	1011 1101	1_11 11_1	
4	1111		

Prime implicants

$$a'c' + a'd' + c'd + ad$$

	$a'c'$ 0_0_	$a'd'$ 0__0	$c'd$ __01	ad 1__1
0000	e	e		

	$a'c'$ 0_0_	$a'd'$ 0__0	$c'd$ _01	ad 1__1
0001	o		o	
0010		E		
0100	e	e		
0101	o		o	
0110		E		
1001			e	e
1011				E
1101			e	e
1111				E

Essential Prime Implicants

$$a'd' + ad$$

Minimal Covers:

$$a'd' + ad + a'c'$$

$$a'd' + ad + c'd$$

C

$$F = \Sigma(1, 2, 3, 6, 7, 8, 9, 12, 14)$$

✓ Answer

1s	Binary	2	3
0			
1	0001 0010 1000	00_1 _001 001_ 0_10 100_ 1_00	00_1 _001 0_1_ 100_ 1_00
2	0011 0110 1001 1100	0_11 011_ _110 11_0	_110 11_0
3	0111 1110		
4			

Prime implicants

$$a'b'd + b'c'd + a'c + ab'c' + ac'd' + bcd' + abd'$$

	$a'b'd$ 00_1	$b'c'd$ _001	$a'c$ 0_1_	$ab'c'$ 100_	$ac'd'$ 1_00	bcd' _110	abd' 11_0
0001	u	P					
0010			E				
1000				o	o		
0011	e		e				
0110			e			e	
1001		p		p			
1100					p		p
0111			E				
1110						u	P

Essential Prime Implicants

$a'c$

Minimal Covers:

$a'c + b'c'd + abd' + ab'c'$

$a'c + b'c'd + abd' + ac'd'$

$a'c + a'b'd + abd' + ab'c'$

$a'c + b'c'd + bcd' + ac'd'$

d

$F = \Sigma(0, 2, 4, 5, 10, 11, 13, 15)$

✓ Answer

1s	Binary	2	3
0	0000	00_0 0_00	00_0 0_00
1	0010 0100	010_ _100	_10_
2	0101 1100	_101 110_	
3	1011 1101	1_11 11_1	1_11 11_1
4	1111		

Prime implicants

$a'b'd' + a'c'd' + bc' + acd + abd$

	$a'b'd'$ 00_0	$a'c'd'$ 0_00	bc' _10_	acd 1_11	abd 11_1
0000	e	e			

	$a'b'd'$ 00_0	$a'c'd'$ 0_00	bc' _10_	acd 1_11	abd 11_1
0010	E				
0100		e	e		
0101			E		
1100			E		
1011				E	
1101			e		u
1111				e	e

Essential Prime Implicants and minimal cover

$$a'b'd' + bc' + acd$$

4

For the given function:

$$F(w, x, y, z) = \Sigma(0, 1, 2, 6, 7, 8, 9, 12, 13) + d(5, 10, 14)$$

a

Using Karnaugh map, find all prime implicants?

✓ Answer

$wx \backslash yz$	00	01	11	10
00	1	1	0	1
01	0	d	1	1
11	1	1	0	d
10	1	1	0	d

Prime Implicants:

$$y'z + yz' + wy' + x'y' + w'xy + w'x'z'$$

b

Which prime implicants are essential?

✓ Answer

	$y'z$ _01	yz' _10	wy' 1_0_	$x'y'$ _00_	$w'xy$ 011_	$w'x'z'$ 00_0
0000				x		x
0001	x			x		



e

Find a function H such that $F = (y' + z')(w' + z')H$ if such H exists?

✓ **Answer**

Not possible as the value 7 is already turned off.